Comprehensive Analysis of Evolutionary Optimization in Water Distribution Systems: State-of-the-Art and Benchmark Instances

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1. Introduction

The optimization of water distribution networks (WDNs) is a fundamental discrete non-linear optimization problem in civil engineering. The primary goal is to balance massive capital investment with strict hydraulic reliability. In the paradigm of *Algorithm Engineering*, transitioning from a conceptual model to a robust solver requires tackling a notoriously difficult search space.

Since pipes must be selected from a discrete set of commercially available diameters, the search space size is defined as *nsnp*, where *ns* is the number of available pipe sizes and *np* is the number of pipes in the network. For complex networks, this space grows exponentially (e.g., 1.23×1027 for the GoYang network), rendering exact deterministic approaches (like standard Linear Programming) inefficient due to their tendency to get trapped in local minima. Consequently, the State-of-the-Art (SOTA) heavily relies on metaheuristic evolutionary approaches, particularly Genetic Algorithms (GAs).

2. Mathematical Formulation and Hydraulic Constraints

The optimization algorithm aims to find a configuration of pipe diameters that minimizes the total network cost, acting as the objective function:

$$C_T = \sum_{i=1}^{N_p} C_i(D_i) \cdot L_i$$

where *CT* is the total cost, *Np* is the total number of pipes, *Ci*(*Di*) is the cost per unit length for pipe *i* with diameter *Di*, and *Li* is the pipe length.

To evaluate a candidate solution, the algorithm must interface with a hydraulic solver (such as EPANET) to verify implicit physical constraints. These constraints dictate that the fluid mechanics laws are satisfied:

1. **Conservation of Mass (Continuity Equation):** At each junction node, the algebraic sum of flow rates must be zero:

$$\sum_{i=1}^{n} q_i = 0$$

where *qi* represents the flow into and out of the node.

2. **Conservation of Energy:** The total head loss around any closed loop must be exactly zero:

$$\sum_{i=1}^{m} h_i = 0$$

where *hi* is the head loss in each pipe, and *m* is the number of pipes in the loop.

3. **Friction Head Loss (Hazen-Williams Equation):** The energy loss due to friction in each pipe is modeled as:

$$h_f = 4.72 \cdot C^{-1.85} \cdot Q^{1.85} \cdot D^{-4.87} \cdot L$$

where *Q* is the flow rate, *C* is the Hazen-Williams roughness coefficient, and *D* is the inner pipe diameter.

3. State-of-the-Art (SOTA) Evolutionary Approaches

The SOTA in least-cost WDN design uses specialized mechanisms to handle the discrete search space and severe hydraulic constraints.

3.1 The SOP-WDN Framework and Algorithmic Mechanics

Recent advancements are best illustrated by the Smart Optimization Program for Water Distribution Networks (SOP-WDN), which couples a GA with the EPANET solver. It addresses the problem through carefully engineered operators:

- **Encoding and Redundancy:** Solutions are encoded using Reflected Binary Code (Gray code) to ensure successive values differ by only one bit. Redundant binary codes are handled by linearly scaling the probability of allocation, where the largest commercial pipe is given twice the allocation chance of the smallest, preserving valuable genetic diversity.
- **Selection Pressure:** The algorithm utilizes a power-law-based probability selection method to balance exploration and exploitation. The selection probability *P*(*i*) for an individual ranked *i* is:

$$P(i) = \frac{i^{-\tau}}{\sum_{k=1}^{n} k^{-\tau}}$$

where empirical tuning shows $\tau \in [1.1, 1.2]$ provides the optimal selection pressure.

- **Search Operators:** The algorithm applies a k-point crossover (where $k \approx 0.8 \times Np$) with an 85–90% probability. To prevent premature convergence, a bit-flip mutation is applied at a rate of 4–6%, combined with an elitism operator that preserves the 2 best solutions per generation.
- **Penalty Functions (Fitness Evaluation):** The fitness of a solution is the reciprocal of the total cost multiplied by severe penalties for explicit constraint violations. The Pressure Penalty (PP) and Velocity Penalty (VP) are computed as:

$$PP = 1 + \sum_{j=1}^{N_n} |TP - P_j| \cdot PP_1 + \sum_{j=1}^{N_n} |TP - P_j| \cdot PP_2$$

$$VP = 1 + \sum_{i=1}^{N_p} |TV - V_i| \cdot VP_1 + \sum_{i=1}^{N_p} |TV - V_i| \cdot VP_2$$

where TP and TV are the target pressure and velocity, and $PP1, PP2, VP1, VP2$ are severity coefficients depending on whether the violation is above or below the targeted bounds.

3.2 Competing SOTA Metaheuristic

While GAs are highly effective, the current SOTA encompasses other stochastic algorithms that compete in finding near-optimal costs on standard benchmarks.

Table 1: Comparison of SOTA Metaheuristic Algorithms for WDN Optimization

Algorithm	**Main Characteristics & SOTA Advantage**
Genetic Algorithm (GA)	Based on the mechanics of natural selection and genetics. Its major advantage is the ability to **generate multiple alternative solutions that are both practical and exceptionally close to the optimum**.
Shuffled Complex Evolution (SCE)	A robust metaheuristic demonstrated to be a highly viable alternative to classical optimization algorithms, primarily due to its **significantly faster computational speed**.
Harmony Search (HS)	An algorithm that has consistently obtained **comparable and highly competitive results** on benchmark instances, proving to be an effective tool for the optimal design of water networks.
Shuffled Frog-Leaping Algorithm (SFLA)	A memetic metaheuristic specifically recognized in recent literature for successfully addressing the discrete optimization of pipe diameters.
Particle Swarm Optimization (PSO)	Modified and hybrid versions of PSO have been actively and successfully utilized to achieve cost minimization in complex distribution networks.

4. Benchmark Instances for WDN Optimization

Benchmark water distribution networks provide a common, standardized testbed for evaluating the performance, reliability, and computational efficiency of newly developed metaheuristic algorithms. To prove their significance and robustness, evolutionary algorithms like the Genetic Algorithm (GA) used in the SOP-WDN framework are applied to these classical WDN problems and compared with existing literature. For this study, we focus on three core benchmark instances, each presenting distinct topological characteristics and hydraulic challenges:

- **Two-Loop Network (TLN):** A synthetic network originally introduced by Alperovitz and Shamir, consisting of 7 nodes and 8 pipelines. Fed by gravity from a single reservoir with an elevation of 210 m, it serves as a fundamental small-scale testbed requiring a minimum head of 30 m at each node.

- **Hanoi Network (HAN):** A real-world, medium-scale trunk system located in Vietnam, first presented by Fujiwara and Kang. It comprises 32 nodes, 34 pipes, and 3 loops, and is fed by gravity from a reservoir with a 100 m fixed head. The network requires a minimum pressure limitation of 30 m.

- **GoYang Network (GOY):** A highly complex network located in South Korea, introduced by Kim et al., characterized by a high loop density. It consists of 22 nodes, 30 pipes, and 9 loops, and is fed from a 71 m constant head reservoir using a single fixed pump. The minimum head limitation for this dense network is 15 m above ground level.

Table 2: Technical Specifications of Core WDN Benchmarks

Network Name	Nodes	Pipes	Sources	Loops	Search Space	Characteristics
Two-Loop (TLN)	7	8	1	2	1.67×10^{7}	Small synthetic network
Hanoi (HAN)	32	34	1	3	2.86×10^{26}	Medium-scale trunk system
GoYang (GOY)	22	30	1	9	1.23×10^{27}	High loop density

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